

LAB 4: PROBABILITY DISTRIBUTIONS

⇒ "RANDOM VARIABLE" - VALUE RESULTS FROM MEASUREMENTS OF A QUANTITY SUBJECT TO RANDOM VARIATIONS

Types - Discrete

- Continuous

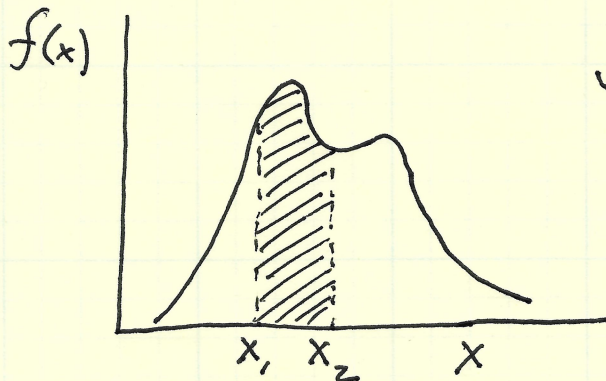
"INDEPENDENT & IDENTICALLY DISTRIBUTED" (IID)

- VARIABLES DRAWN FROM SAME DISTRIBUTION, THAT ARE INDEPENDENT

↳ IID RV if $p(x, y) = p(x)p(y)$

⇒ PROBABILITY & PROBABILITY DENSITY

IN A CONTINUOUS DISTRIBUTION, THE PROBABILITY OF MEASURING ANY PARTICULAR VALUE IS ZERO! THE BEST WE CAN DO IS DEFINE A "PROBABILITY DENSITY FUNCTION" (PDF) SUCH THAT...



$$\int_{x_1}^{x_2} f(x) dx = \text{Probability of measuring } x_1 < X < x_2$$

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

⇒ Descriptive Statistics (Location, scale, shape of PDF)

- Mean
(expectation value) $\mu = E(x) = \int_{-\infty}^{\infty} x f(x) dx$

- Variance
(STD. DEV.)² $\sigma^2 = V = \int_{-\infty}^{\infty} (x - \mu)^2 f(x) dx$

- Skewness
(asymmetry) $\Sigma' = \int_{-\infty}^{\infty} \left(\frac{x - \mu}{\sigma} \right)^3 f(x) dx$

- Kurtosis
(HEAVY-TAILEDNESS) $K = \underbrace{\int_{-\infty}^{\infty} \left(\frac{x - \mu}{\sigma} \right)^4 f(x) dx}_{\text{excess kurtosis}} - 3$
= ϕ for Normal/Gaussian

Collectively "moments" of distribution $f(x)$

- Absolute Deviation
about d $\delta = \int_{-\infty}^{\infty} |x - d| f(x) dx$

- Mode
(Most Probable Value) $\left. \frac{df(x)}{dx} \right|_{x=x_m} = \phi$

$\Rightarrow$ ESTIMATES OF DESCRIPTIVE STATISTICS FROM DATA

Mean $\mu = \int_{-\infty}^{\infty} x f(x) dx \rightarrow \bar{X} = \frac{1}{N} \sum_{i=1}^N x_i$

Std Dev $\sigma = \sqrt{\int_{-\infty}^{\infty} (x-\mu)^2 f(x) dx} \rightarrow S = \sqrt{\frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{X})^2}$

↑
"Bessel's Correction"
(lose a degree of freedom
when deriving $\bar{X}$ from data!)

Uncertainties $\sigma_{\bar{X}} = \frac{S}{\sqrt{N}}$

$$\sigma_S = \frac{S}{\sqrt{2(N-1)}}$$

* * *

LAB 4: Exponential Distribution

$$f(x) = \frac{1}{\beta} e^{-x/\beta} \leftarrow \text{PYTHON CONVENTION!}$$

$$\text{EQUALLY } f(x) = \beta e^{-\beta x}$$

$$\int_0^{\infty} \frac{1}{\beta} e^{-x/\beta} dx = \frac{1}{\beta} [-\beta e^{-x/\beta}]_0^{\infty} = 1 \quad \checkmark$$

< NOTEBOOK : rv-gallery.ipynb >